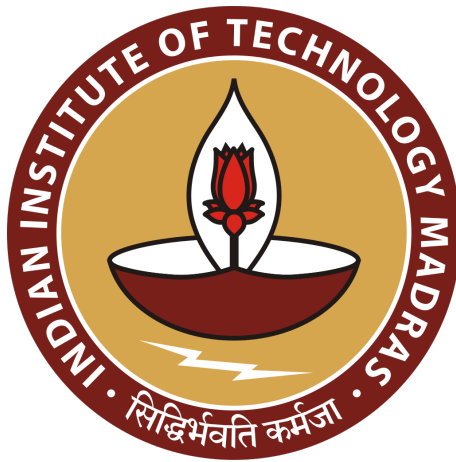


Lab Assignment - 6

Determining Conservation Scores



BT3040 - Bioinformatics

Even Semester 2026

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Question 1

1. Open **Windows PowerShell** in the directory that contains the input files and where the output files are intended to be stored. This simplifies file transfer operations later.
2. Log into the remote Linux system using the **ssh** command, and the entered the password when prompted:

```
ssh <username>@<server_ip>          (ssh bt3040@10.21.49.164)
```

3. Create a directory on the remote server using your roll number:

```
mkdir <roll_number>          (mkdir BE23B016)
```

4. Change your working directory to the newly created directory:

```
cd <roll_number>          (cd BE23B016)
```

5. Use the **pwd** (Present working directory command) in the remote server to get the path of your file in the remote server.

```
pwd          (it gives an output -> /home/bt3040/BE23B016)
```

6. Open another terminal window on your local system. This terminal is used to transfer files between the local system and the remote server.
7. Upload the input alignment files to the server using the **scp** command (Secure Copy Protocol):

```
scp <local_file> <username>@<server_ip>:<remote_directory>
```

```
(scp clustalo* bt3040@10.21.49.164:~/BE23B016)
```

NOTE: ~ here is a path representing /home/bt3040

8. After the transfer is complete, verify it's presence using the **ls** command on the remote server.
9. To remove the header line in clustal alignment file use the nano command.

```
nano <file_name>
```

Use the **Delete** key to remove the header and exit using **Ctrl+X**.

10. Run **al2co** using the command format given below.

```
al2co -i <input_file_name> -o <output_file_name> -f<0/1/2> -c<0/1/2> -g1 -n<T/F>
```

Ensure that the output file has an extension such as **.txt**.

11. Verify the generated output file using:

```
cat <output_file_name>
```

12. Repeat the command with different parameter combinations as required for Question 1

13. After generating all 10 required output files (5 for Set-1 and 5 for Set-2), go to the local terminal opened earlier
14. In the local terminal download the files from the server using:

```
scp -r <username>@<server_ip>:<remote_directory> .
```

```
(scp -r bt3040@10.21.49.164:~/BE23B016 .)
```

The . represents the current local directory.

15. Finally, log out of the SSH session using **Ctrl+D**:

NOTE: Unweighted frequency, sum of pairs score was calculated using the web server to use BLOSUM62 as the scoring matrix instead of the default identity matrix [Part iii) has 2 images attached].

The output files were obtained and the results are shown below:

1	-	0.000
2	S	-0.305
3	L	0.000
4	S	0.000
5	D	-1.121
6	K	-1.160
7	D	0.000
8	K	0.000
9	A	-1.169
10	A	-0.305
11	V	-0.586
12	R	-0.305
13	A	-1.034
14	L	-1.160
15	M	-0.305
16	S	-0.916
17	K	0.000
18	I	-0.689
19	G	-0.305
20	K	-1.034
21	S	-0.886
22	A	-0.474
23	D	-1.034
24	A	-0.908
25	I	-0.886
26	G	0.000
27	H	-0.886
28	D	-0.305
29	A	-0.305
30	L	0.000
31	S	-0.886
32	R	0.000
33	M	-0.305
34	I	-0.305
35	V	-1.205
36	V	-1.367
37	Y	-0.474
38	P	0.000
39	Q	-0.474
40	T	0.000

Figure 1.1: i) Unweighted frequency and entropy-based measure, Set -1

1	H	0.000
2	A	0.000
3	S	-0.974
4	K	-1.494
5	P	-0.377
6	Q	-0.756
7	P	-0.377
8	I	-0.736
9	A	-0.377
10	A	-0.377
11	A	-0.377
12	H	0.000
13	H	-0.349
14	K	0.000
15	C	-0.937
16	H	0.000
17	G	-0.349
18	S	-1.311
19	E	-1.303
20	S	-1.523
21	L	-1.581
22	L	-1.061
23	V	-1.677
24	P	-1.003
25	L	-0.687
26	I	-0.965
27	E	-1.725
28	T	-1.003
29	L	-0.349
30	H	-0.349
31	A	-1.003
32	A	-0.684
33	T	-1.889
34	F	-1.581
35	D	-1.149
36	H	-1.677
37	D	-1.523
38	-	-0.693
39	-	-0.693
40	V	-0.687

Figure 1.2: i) Unweighted frequency and entropy-based measure, Set -2

1	-	1.021
2	S	0.865
3	L	0.909
4	S	0.953
5	D	0.527
6	K	0.561
7	D	0.973
8	K	0.956
9	A	0.541
10	A	0.917
11	V	0.744
12	R	0.871
13	A	0.567
14	L	0.559
15	M	0.943
16	S	0.612
17	K	0.956
18	I	0.691
19	G	0.886
20	K	0.613
21	S	0.713
22	A	0.740
23	D	0.624
24	A	0.665
25	I	0.757
26	G	0.977
27	H	0.650
28	D	0.922
29	A	0.817
30	L	0.809
31	S	0.749
32	R	1.015
33	M	0.933
34	I	0.904
35	V	0.470
36	V	0.472
37	Y	0.830
38	P	0.993
39	Q	0.821
40	T	0.977

Figure 1.3: ii) Unweighted frequency and variance-based measure, Set -1

1	H	1.021
2	A	0.914
3	S	0.648
4	K	0.426
5	P	0.882
6	Q	0.726
7	P	0.884
8	I	0.769
9	A	0.881
10	A	0.812
11	A	0.812
12	H	0.995
13	H	0.909
14	K	0.962
15	C	0.679
16	H	0.995
17	G	0.835
18	S	0.510
19	E	0.553
20	S	0.401
21	L	0.481
22	L	0.526
23	V	0.317
24	P	0.640
25	L	0.664
26	I	0.550
27	E	0.362
28	T	0.663
29	L	0.859
30	H	0.888
31	A	0.591
32	A	0.691
33	T	0.385
34	F	0.426
35	D	0.604
36	H	0.372
37	D	0.429
38	-	0.621
39	-	0.661
40	V	0.645

Figure 1.4: ii) Unweighted frequency and variance-based measure, Set -2

1	-	1.000
2	S	0.835
3	L	1.000
4	S	1.000
5	D	0.388
6	K	0.438
7	D	1.000
8	K	1.000
9	A	0.372
10	A	0.835
11	V	0.603
12	R	0.835
13	A	0.455
14	L	0.438
15	W	0.835
16	S	0.438
17	K	1.000
18	T	0.504
19	G	0.835
20	K	0.455
21	S	0.554
22	A	0.702
23	D	0.455
24	A	0.471
25	I	0.554
26	G	1.000
27	N	0.554
28	D	0.835
29	A	0.835
30	L	1.000
31	S	0.554
32	R	1.000
33	M	0.835
34	I	0.835
35	V	0.355
36	V	0.386
37	Y	0.702
38	P	1.000
39	Q	0.702
40	T	1.000

Figure 1.5: iii) Unweighted frequency and sum of pairs-based measure, Set -1 (Identity)

1	M	1.000
2	A	1.000
3	S	0.406
4	K	0.250
5	P	0.781
6	Q	0.594
7	P	0.781
8	I	0.594
9	A	0.781
10	A	0.781
11	A	0.781
12	N	1.000
13	H	0.802
14	K	1.000
15	C	0.432
16	N	1.000
17	G	0.802
18	S	0.284
19	E	0.358
20	S	0.235
21	L	0.259
22	L	0.358
23	V	0.210
24	P	0.481
25	L	0.586
26	I	0.407
27	E	0.185
28	T	0.481
29	L	0.882
30	N	0.802
31	A	0.481
32	A	0.430
33	T	0.160
34	F	0.259
35	D	0.383
36	H	0.210
37	D	0.235
38	-	0.580
39	-	0.580
40	V	0.586

Figure 1.6: iii) Unweighted frequency and sum of pairs-based measure, Set -2 (Identity)

1	:	5.000
2	S	3.000
3	L	4.000
4	S	4.000
5	D	1.149
6	K	1.157
7	D	6.000
8	K	5.000
9	A	1.868
10	A	4.661
11	V	3.603
12	R	4.504
13	A	1.909
14	L	1.521
15	W	9.306
16	S	2.264
17	K	5.000
18	I	3.504
19	G	4.992
20	K	2.445
21	S	4.372
22	A	2.876
23	D	2.240
24	A	2.455
25	I	4.314
26	G	6.000
27	N	1.826
28	D	4.512
29	A	3.347
30	L	4.000
31	S	3.298
32	R	5.000
33	M	4.008
34	I	4.992
35	V	0.959
36	V	1.132
37	Y	5.140
38	P	7.000
39	Q	2.215
40	T	5.000

Figure 1.7: iii) Unweighted frequency and sum of pairs-based measure, Set -1 (BLOSUM62)

1	M	5.000
2	A	4.000
3	S	1.328
4	K	1.109
5	P	3.500
6	Q	2.938
7	P	3.828
8	I	3.172
9	A	3.125
10	A	4.656
11	A	4.656
12	N	6.000
13	M	8.963
14	K	5.000
15	C	2.395
16	N	6.000
17	G	4.000
18	S	0.790
19	E	1.568
20	S	0.901
21	L	0.519
22	L	2.667
23	V	0.198
24	P	2.025
25	L	3.012
26	I	2.716
27	E	0.136
28	T	1.615
29	L	3.235
30	N	4.988
31	A	1.951
32	A	2.716
33	T	0.383
34	F	0.457
35	D	1.541
36	H	0.383
37	D	1.259
38	:	1.500
39	:	1.500
40	V	2.222

Figure 1.8: iii) Unweighted frequency and sum of pairs-based measure, Set -2 (BLOSUM62)

1	-	1.919
2	S	0.729
3	L	0.916
4	S	0.951
5	D	0.492
6	K	0.439
7	D	0.972
8	K	0.950
9	A	0.465
10	A	0.770
11	V	0.763
12	R	0.744
13	A	0.566
14	L	0.415
15	M	0.888
16	S	0.604
17	K	0.950
18	I	0.719
19	G	0.831
20	K	0.582
21	S	0.627
22	A	0.767
23	D	0.517
24	A	0.593
25	I	0.679
26	G	0.977
27	N	0.561
28	D	0.784
29	A	0.768
30	L	0.916
31	S	0.628
32	R	1.011
33	M	0.944
34	I	0.780
35	V	0.373
36	V	0.386
37	Y	0.718
38	P	0.593
39	Q	0.715
40	T	0.979

Figure 1.9: iv) Weighted frequency and variance-based measure, Set -1

1	M	1.021
2	A	0.907
3	S	0.656
4	K	0.365
5	P	0.827
6	Q	0.641
7	P	0.830
8	I	0.713
9	A	0.739
10	A	0.754
11	A	0.754
12	N	0.994
13	M	0.876
14	K	0.966
15	C	0.639
16	N	0.994
17	G	0.810
18	S	0.530
19	E	0.515
20	S	0.372
21	L	0.482
22	L	0.514
23	V	0.270
24	P	0.597
25	L	0.656
26	I	0.557
27	E	0.393
28	T	0.595
29	L	0.826
30	N	0.885
31	A	0.549
32	A	0.676
33	T	0.282
34	F	0.364
35	D	0.553
36	H	0.248
37	D	0.421
38	-	0.621
39	-	0.662
40	V	0.684

Figure 1.10: iv) Weighted frequency and variance-based measure, Set -2

1	-	0.937
2	S	0.246
3	L	0.937
4	S	0.937
5	D	-1.684
6	K	-1.692
7	D	0.937
8	K	0.937
9	A	-1.712
10	A	0.246
11	V	-0.391
12	R	0.246
13	A	-1.486
14	L	-1.692
15	W	0.246
16	S	-1.141
17	K	0.937
18	I	-0.625
19	G	0.246
20	K	-1.486
21	S	-1.071
22	A	-0.138
23	D	-1.486
24	A	-1.121
25	I	-1.071
26	G	0.937
27	N	-1.071
28	D	0.246
29	A	0.246
30	L	0.937
31	S	-1.071
32	R	0.937
33	M	0.246
34	I	0.246
35	V	-1.998
36	V	-2.162
37	Y	-0.138
38	P	0.937
39	Q	-0.138
40	T	0.937

Figure 1.11: v) Unweighted frequency and entropy-based measure, Set -1 (Normalized)

1	M	1.169
2	A	1.169
3	S	-0.611
4	K	-1.561
5	P	0.481
6	Q	-0.175
7	P	0.481
8	I	-0.175
9	A	0.481
10	A	0.481
11	A	0.481
12	H	1.169
13	W	0.532
14	K	1.169
15	C	-0.543
16	H	1.169
17	G	0.532
18	S	-1.226
19	E	-1.212
20	S	-1.614
21	L	-1.720
22	L	-0.769
23	V	-1.895
24	P	-0.663
25	L	-0.086
26	I	-0.594
27	E	-2.001
28	T	-0.663
29	L	0.532
30	H	0.532
31	A	-0.663
32	A	-0.080
33	T	-2.283
34	F	-1.720
35	D	-0.931
36	H	-1.895
37	D	-1.614
38	-	-0.098
39	-	-0.098
40	V	-0.086

Figure 1.12: v) Unweighted frequency and entropy-based measure, Set -2 (Normalized)

NOTE: The files are available in the google drive attached in the report

Question 2

Top 10 residues with the highest conservation scores in method (i) are given below:

S.No	Set-1 Residue position	Set-1 Score	Set-2 Residue position	Set-2 Score
1	3, L	0.0	1, M	0.0
2	4, S	0.0	2, A	0.0
3	7, D	0.0	12, N	0.0
4	8, K	0.0	14, K	0.0
5	17, K	0.0	16, N	0.0
6	26, G	0.0	43, V	0.0
7	30, L	0.0	68, Q	0.0
8	32, R	0.0	69, N	0.0
9	38, P	0.0	76, G	0.0
10	40, T	0.0	77, A	0.0

Top 10 residues with the lowest conservation scores in method (i) are given below:

S.No	Set-1 Residue position	Set-1 Score	Set-2 Residue position	Set-2 Score
1	117, K	-1.846	33, T	-1.889
2	113, T	-1.72	64 Q	-1.831
3	70, L	-1.673	147 A	-1.831
4	73, S	-1.594	27, E	-1.735
5	84, E	-1.414	23, V	-1.677
6	133, G	-1.414	36, H	-1.677
7	36, V	-1.367	49, L	-1.677
8	35, V	-1.295	93, S	-1.677
9	9, A	-1.169	145, R	-1.677
10	14, L	-1.160	196, L	-1.677

Question 3

We first compute the required frequencies for both sets. Since the resulting scores are stored in a large dictionary and are too lengthy to display in full here, a plot of conservation score versus residue position is shown instead. The exact numerical scores can be found in the accompanying code.

```
def unweighted_freq(data):
    distribution = []

    for n in range(data.get_alignment_length()):
        temp, counter = {}, 0
        AAs = "ACDEFGHIKLMNPQRSTVWY"

        for AA in AAs:
            temp[AA] = 0

        for seq_no in range(len(data)):
            if data[seq_no].seq[n].isalpha():
                counter += 1
                temp[data[seq_no].seq[n]] += 1

        for residue in temp.keys():
            temp[residue] = temp[residue]/counter

        distribution.append((n,temp))
    return(distribution)

def general_freq(data):
    frequency_total, total_AA = {}, 0
    AAs = "ACDEFGHIKLMNPQRSTVWY"

    for AA in AAs:
        frequency_total[AA] = 0

    for record in data:
        for AA in record.seq:
            if AA.isalpha():
                frequency_total[AA] +=1
                total_AA += 1

    for AA in frequency_total.keys():
        frequency_total[AA] = (
            frequency_total[AA]/(data.get_alignment_length()*len(data))
        )

    return(frequency_total)

freq_1 = unweighted_freq(set_1)
freq_2 = unweighted_freq(set_2)
total_freq_1 = general_freq(set_1)
total_freq_2 = general_freq(set_2)
```

```

def entropy_based_score(freq):
    scores = {}

    for n in range(len(freq)):

        residue, frequency = freq[n]
        score = 0

        for AA in frequency.keys():
            if frequency[AA] !=0:
                score += frequency[AA] * (math.log(frequency[AA]))

        scores[residue+1] = round(score,3)

    return(scores)

ebs_1 = entropy_based_score(freq_1)
ebs_2 = entropy_based_score(freq_2)

```

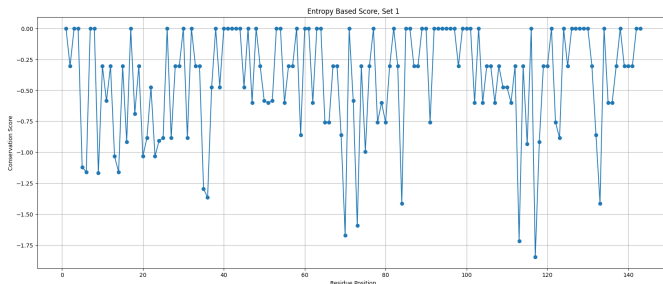


Figure 3.1: Unweighted frequency and entropy-based measure, Set -1

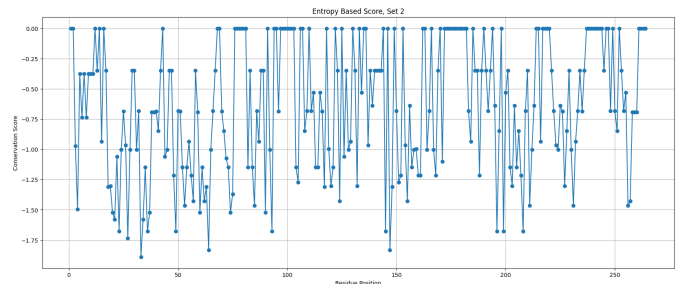


Figure 3.2: Unweighted frequency and entropy-based measure, Set -2

```

def variance_based_score(freq, freq_total):
    scores = {}

    for n in range(len(freq)):

        residue, frequency = freq[n]
        score = 0

        for AA in frequency.keys():
            score += (frequency[AA]-freq_total[AA])**2

        score = score**0.5

        scores[residue+1] = round(score,3)

    return(scores)

vbs_1 = variance_based_score(freq_1,total_freq_1)
vbs_2 = variance_based_score(freq_2,total_freq_2)

```

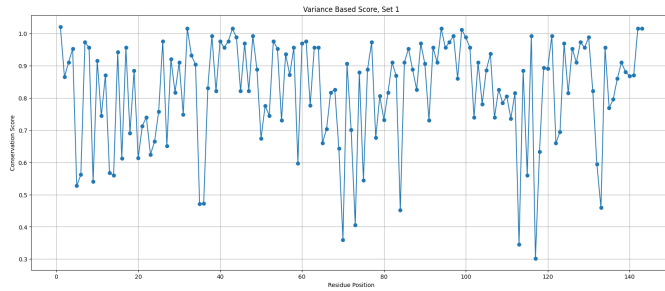


Figure 3.3: Unweighted frequency and variance-based measure, Set -1

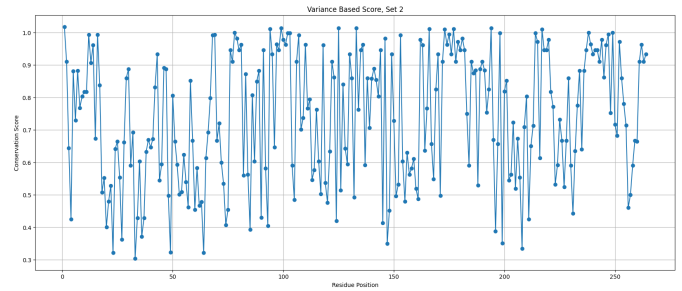


Figure 3.4: Unweighted frequency and variance-based measure, Set -2

```
def sop_based_score_class(freq):

    blosum62 = substitution_matrices.load("BLOSUM62")
    scores = {}

    for n in range(len(freq)):

        residue, frequency = freq[n]
        AAs = "ACDEFGHIKLMNPQRSTVWY"
        score = 0

        for AA1 in AAs:
            for AA2 in AAs:

                score += (frequency[AA1] * frequency[AA2] * blosum62[AA1, AA2])**2

        score = score**0.5

        scores[residue+1] = round(score,3)

    return(scores)

sop_1_class = sop_based_score_class(freq_1)
sop_2_class = sop_based_score_class(freq_2)
```

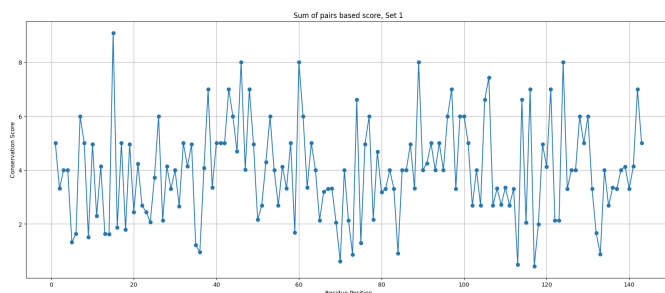


Figure 3.5: Unweighted frequency and sum of pairs based measure, Set -1

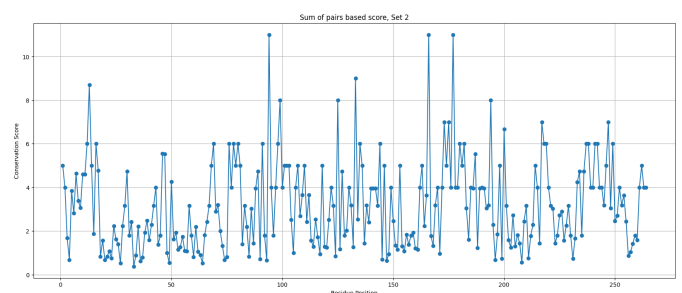


Figure 3.6: Unweighted frequency and sum of pairs based measure, Set -2

Question 4

Once again since the resulting scores are stored in a large dictionary and are too lengthy to display in full here, a plot of conservation score versus residue position is shown instead. here the regions with different residue scores are highlighted as well, with values tabulated below.

```
# Reading all the required MSA files
set_1_clustal = AlignIO.read("/content/drive/MyDrive/BT3040-Assignments/
Lab Assignment - 6/Clustalo-Set-1.aln-clustal_num", "clustal")

set_1_mafft = AlignIO.read("/content/drive/MyDrive/BT3040-Assignments/
Lab Assignment - 6/Set_1_MAFFT.aln", "clustal")

set_1_muscle = AlignIO.read("/content/drive/MyDrive/BT3040-Assignments/
Lab Assignment - 6/Set_1_MUSCLE.aln-clustalw", "clustal")

set_2_clustal = AlignIO.read("/content/drive/MyDrive/BT3040-Assignments/
Lab Assignment - 6/Clustalo-Set-2.aln-clustal_num", "clustal")

set_2_mafft = AlignIO.read("/content/drive/MyDrive/BT3040-Assignments/
Lab Assignment - 6/Set_2_MAFFT.aln", "clustal")

set_2_muscle = AlignIO.read("/content/drive/MyDrive/BT3040-Assignments/
Lab Assignment - 6/Set_2_MUSCLE.aln-clustalw", "clustal")
```

```
freq_1_clustal = unweighted_freq(set_1_clustal)
freq_1_mafft = unweighted_freq(set_1_mafft)
freq_1_muscle = unweighted_freq(set_1_muscle)

ebs_1_clustal = entropy_based_score(freq_1_clustal)
ebs_1_mafft = entropy_based_score(freq_1_mafft)
ebs_1_muscle = entropy_based_score(freq_1_muscle)
```

The data is loaded in the above section and the plots generated from them are shown below:

Set-1

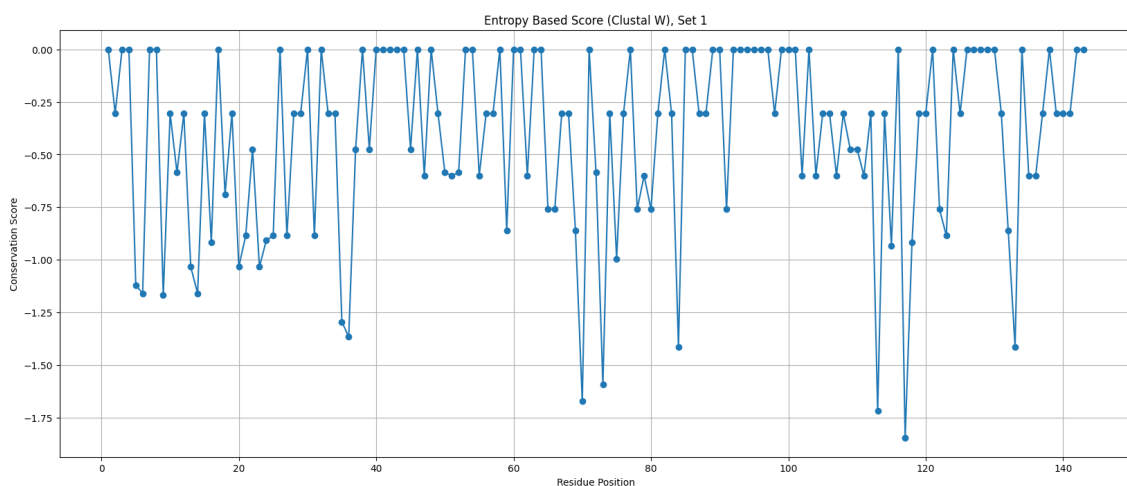


Figure 4.1.1: Clustal-Omega Alignment Scores

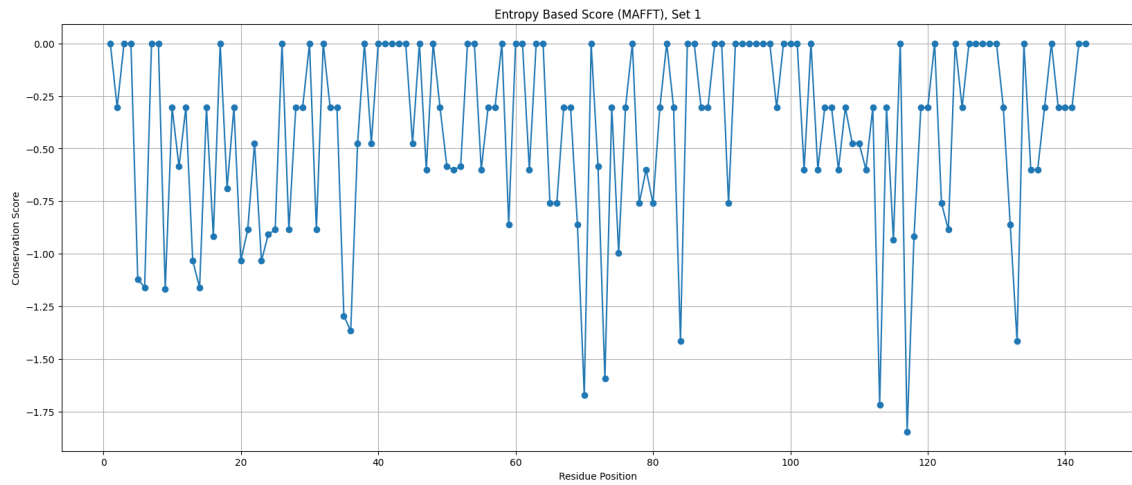


Figure 4.1.2: MAFFT Alignment Scores

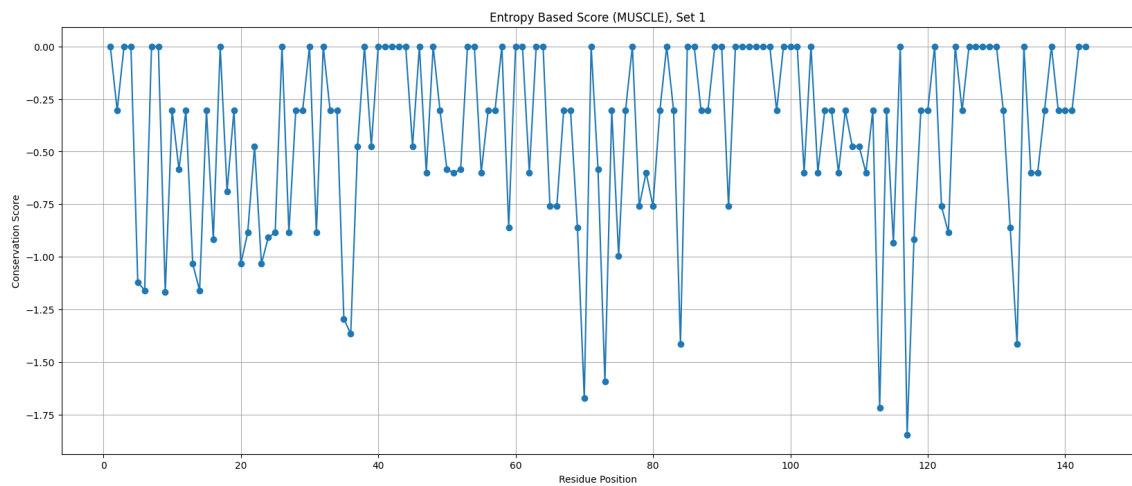


Figure 4.1.3: MUSCLE Alignment Scores

All 3 have the same alignment as indicated by the graphs above (there is no difference)

Set-2

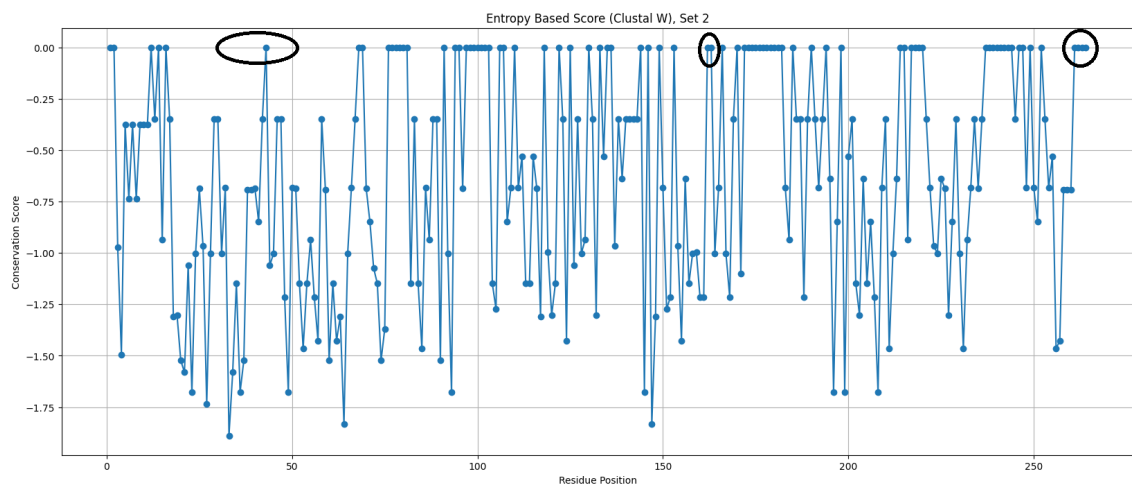


Figure 4.2.1: Clustal-Omega Alignment Scores

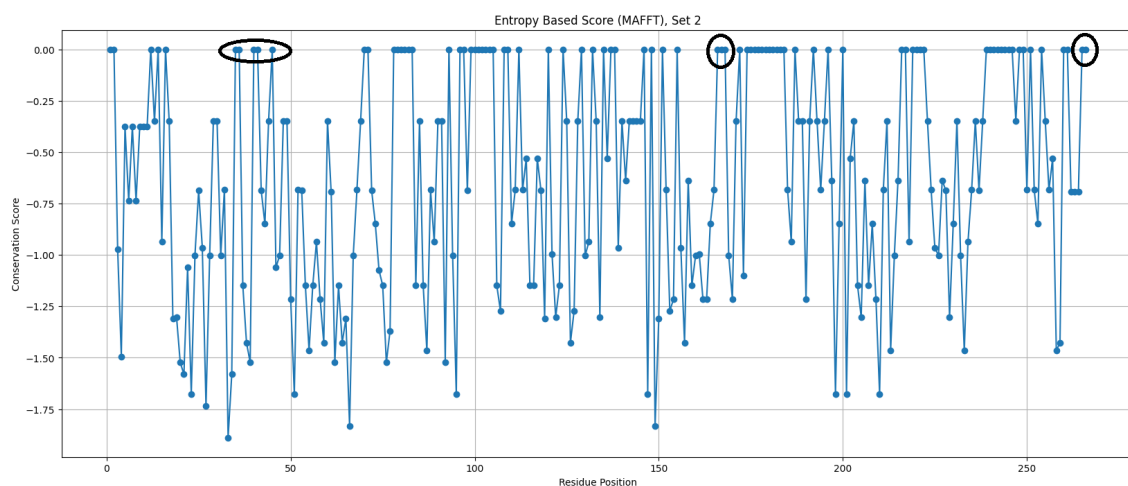


Figure 4.2.2: MAFFT Alignment Scores

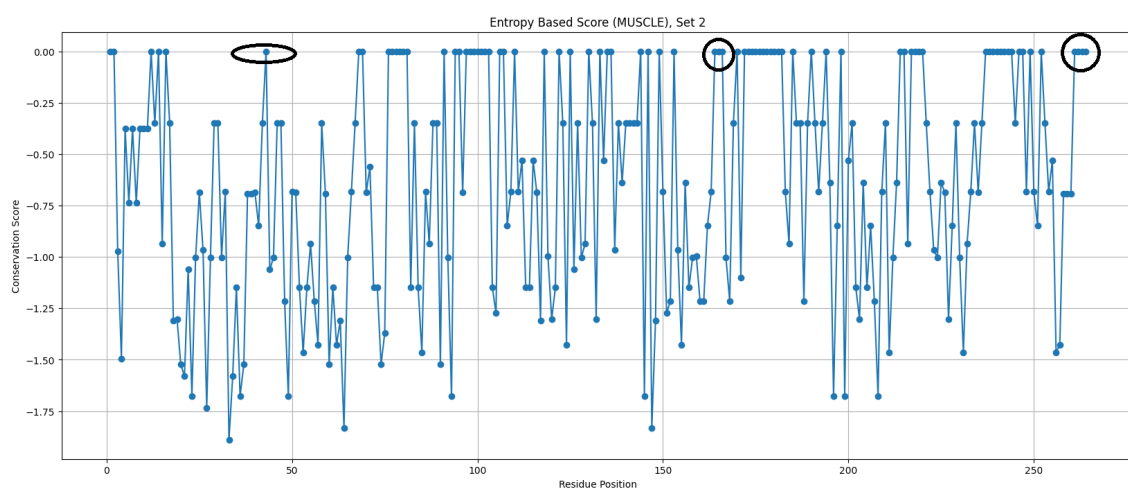


Figure 4.2.3: MUSCLE Alignment Scores

S.No	Residue Position	Clustal-Omega	MAFFT	MUSCLE
1	35	-1.149	0.0	-1.149
2	36	-1.677	0.0	-1.677
3	37	-1.523	-1.149	-1.523
4	38	-0.693	-1.427	-0.693
5	39	-0.693	-1.523	-0.693
6	40	-0.687	0.0	-0.687
7	162	0.0	-1.215	-0.849
8	163	0.0	-1.215	-0.684
9	164	-1.003	-0.849	0.0
10	262	0.0	-0.693	0.0
11	263	0.0	-0.693	0.0
12	264	0.0	-0.693	0.0

As shown in the above graph, in set-2 the alignment score at the above positions are very clearly different.

Question 5

$$C^e(i) = - \sum_{a=1}^{20} f_a(i) \cdot \ln(f_a(i)) \quad i \rightarrow \text{ith residue position}$$

Set-1

i) Position - 9 ; A, S, T, T, A, S, T, T, G, T, T

$$f(A) = \frac{2}{11}; \quad f(S) = \frac{2}{11}; \quad f(G) = \frac{1}{11}; \quad f(T) = \frac{6}{11}$$

$$C^e = 2 \left(\frac{2}{11} \cdot \ln\left(\frac{2}{11}\right) \right) + \frac{1}{11} \ln\left(\frac{1}{11}\right) + \frac{6}{11} \ln\left(\frac{6}{11}\right) = \underline{-1.169}$$

ii) Position - 11; V, V, T, T, V, T, V, V, V, V, V

$$f(T) = \frac{3}{11}; \quad f(V) = \frac{8}{11}$$

$$C^e = \frac{3}{11} \ln\left(\frac{3}{11}\right) + \frac{8}{11} \ln\left(\frac{8}{11}\right) = \underline{-0.586}$$

iii) Position - 20; K, S, G, G, G, G, A, A, G, G, G

$$f(K) = \frac{1}{11}; \quad f(S) = \frac{1}{11}; \quad f(G) = \frac{7}{11}; \quad f(A) = \frac{2}{11}$$

$$C^e = 2 \left(\frac{1}{11} \cdot \ln\left(\frac{1}{11}\right) \right) + \frac{7}{11} \ln\left(\frac{7}{11}\right) + \frac{2}{11} \ln\left(\frac{2}{11}\right) = \underline{-1.034}$$

iv) Position - 22; A, A, A, G, A, G, A, A, A, A, A

$$f(A) = \frac{9}{11}; \quad f(G) = \frac{2}{11}$$

$$C^e = \frac{9}{11} \ln\left(\frac{9}{11}\right) + \frac{2}{11} \ln\left(\frac{2}{11}\right) = \underline{-0.474}$$

v) Position - 30; L, L, L, L, L, L, L, L, L, L, L

$$f(L) = \frac{11}{11} = 1$$

$$C^e = 1 \cdot \ln(1) = \underline{0}$$

Figure 5.1: Set-1, unweighted entropy-based scores manually calculated

Set-2

i) Position - 9; A, -, V, V, V, V, V, V, V

$$f(A) = \frac{1}{8}; \quad f(V) = \frac{7}{8}$$

$$C^e = \frac{1}{8} \cdot \ln\left(\frac{1}{8}\right) + \frac{7}{8} \cdot \ln\left(\frac{7}{8}\right) = \underline{-0.377}$$

ii) Position - 11; A, G, G, G, G, G, G, G, G

$$f(A) = \frac{1}{8}; \quad f(G) = \frac{7}{8}$$

$$C^e = \frac{1}{8} \cdot \ln\left(\frac{1}{8}\right) + \frac{7}{8} \cdot \ln\left(\frac{7}{8}\right) = \underline{-0.377}$$

iii) Position - 20; S, A, Q, D, A, K, Q, K, K

$$f(S) = \frac{1}{9}; \quad f(A) = \frac{2}{9}; \quad f(Q) = \frac{2}{9}; \quad f(D) = \frac{1}{9}; \quad f(K) = \frac{3}{9}$$

$$C^e = 2 \cdot \frac{1}{9} \cdot \ln\left(\frac{1}{9}\right) + 2 \cdot \frac{2}{9} \cdot \ln\left(\frac{2}{9}\right) + \frac{3}{9} \cdot \ln\left(\frac{3}{9}\right) = \underline{-1.523}$$

iv) Position - 22; L, I, I, V, V, I, L, L, L

$$f(L) = \frac{4}{9}; \quad f(I) = \frac{3}{9}; \quad f(V) = \frac{2}{9}$$

$$C^e = \frac{4}{9} \cdot \ln\left(\frac{4}{9}\right) + \frac{3}{9} \cdot \ln\left(\frac{3}{9}\right) + \frac{2}{9} \cdot \ln\left(\frac{2}{9}\right) = \underline{-1.061}$$

v) Position - 20; N, N, N, N, N, S, N, N, N

$$f(N) = \frac{8}{9}; \quad f(S) = \frac{1}{9}$$

$$C^e = \frac{8}{9} \cdot \ln\left(\frac{8}{9}\right) + \frac{1}{9} \cdot \ln\left(\frac{1}{9}\right) = \underline{-0.349}$$

Figure 5.2: Set-2, unweighted entropy-based scores manually calculated

Question 6

At the time of solving the assignment the ConSurf server was down and couldn't be accessed, as such we use **Evolutionary Trace** (the alternative software).

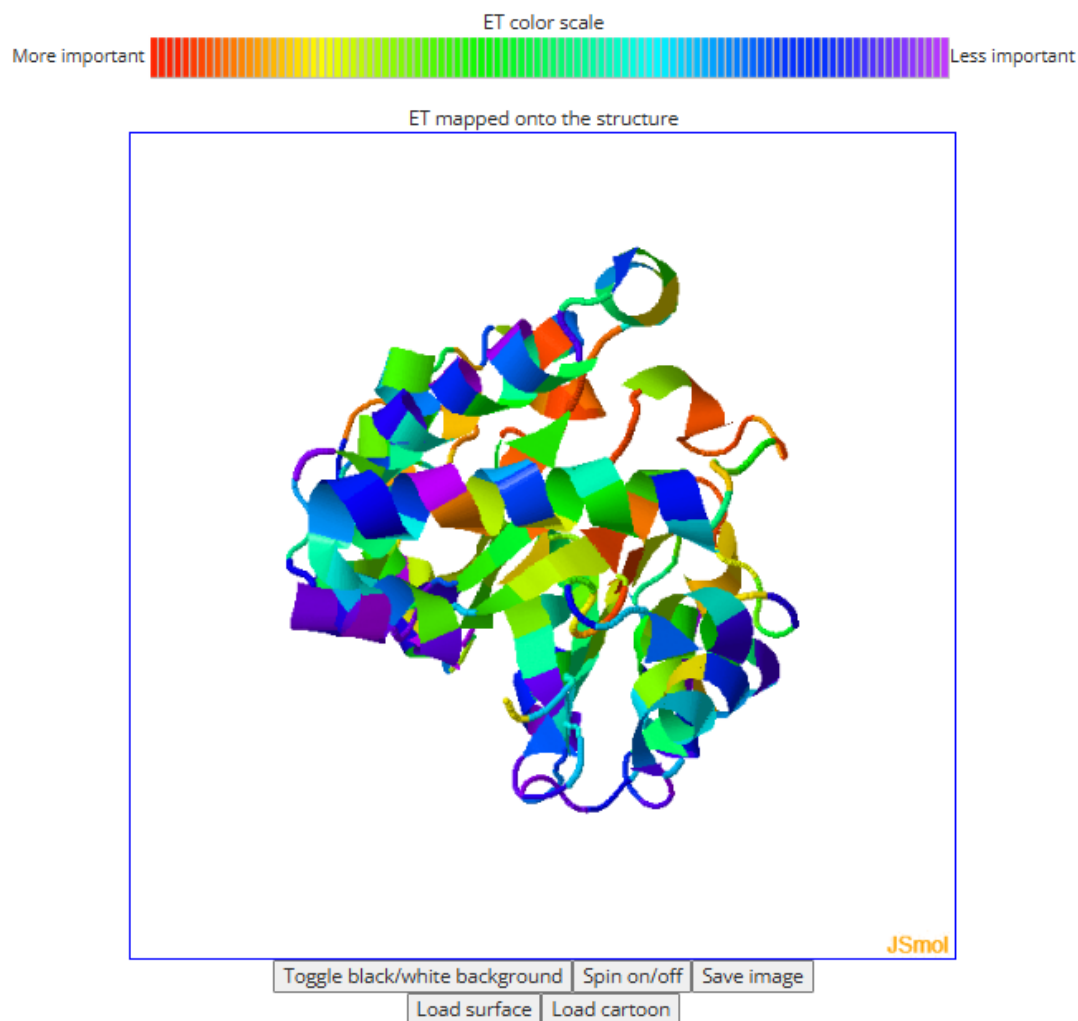


Figure 6.1: 1BTM-A, Evolutionary importance score visualized on the 3D structure of the protein

Click on the amino acid letter and the corresponding residue will be highlighted on the structure as spacefill(spheres)

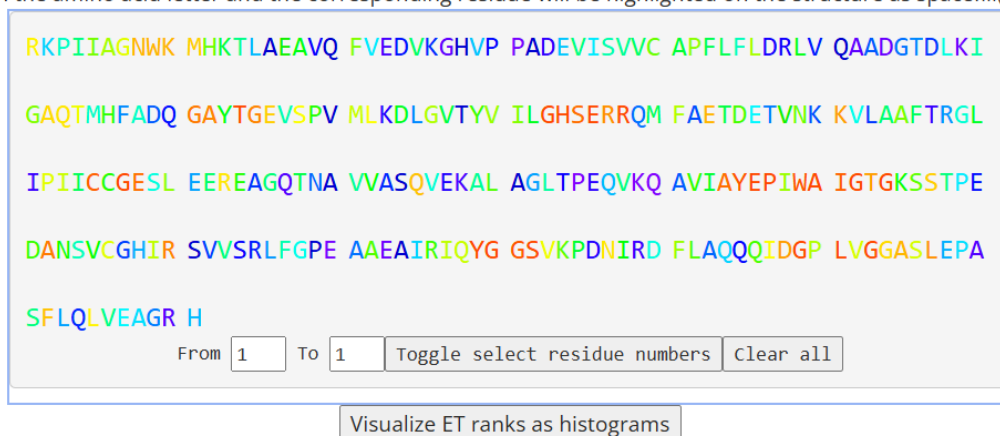


Figure 6.2: 1BTM-A, Evolutionary importance score visualized on the sequence of the protein

The **Evolutionary Trace** software identifies residues that are evolutionarily important, i.e, residues that are typically highly conserved. We use evolutionary importance here as a proxy or stand-in for conservation, based on the reasoning that residues deemed evolutionarily important are likely to be highly conserved, since such regions are critical for the organism's survival and proper functioning.

Regions that appear redder correspond to highly conserved regions, whereas regions that appear bluer regions are more likely to be variable or poorly conserved.

Link to the google drive containing the python code and al2co output files used in this assignment:
<https://drive.google.com/drive/u/0/folders/1tdBibuMKp3GjBlWwak1lSg75JjEWLoyD>

Annotations have been omitted from the submitted PDF for the sake of brevity. The fully annotated version of the code is available in the code folder within the attached Google Drive.